FlowScene: Style-Consistent Indoor Scene Generation with Multimodal Graph Rectified Flow

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Research Questions

- How can we generate style-consistent 3D indoor scenes with fine-grained control over object shapes, textures, and spatial relations?
- How can multimodal graph rectified flow enable collaborative reasoning across objects during scene generation?
- How can we achieve both high-fidelity textured results and scene-level style consistency in graph-conditioned 3D scene generation?

Core Idea

FlowScene addresses the challenge of generating realistic 3D indoor scenes where objects maintain consistent visual style while respecting spatial constraints. The key insight is using a multimodal graph to represent scenes where nodes contain both textual and visual object information, then applying a tri-branch rectified flow model that simultaneously generates layouts, shapes, and textures. The critical trick is the InfoExchangeUnit that allows objects to share information during the denoising process, enabling collaborative reasoning - for example, if one object has a specific texture style, other objects can adapt to maintain consistency across the scene.

Key Formula

Graph Rectified Flow Loss

$$L_{GRF} = E_{D,C,t}\left[\|\Theta_D(D_t, C_t, t) - v\|_2^2\right]$$

L_GRF is the training loss, D_t is noisy data at timestep t, C_t are time-dependent graph-based conditions from InfoExchangeUnit, Θ_D is the denoiser network, v is the target velocity field (D_1 - D_0). This loss trains the model to predict velocity fields while incorporating inter-object constraints through graph-based conditioning.

Key Figure

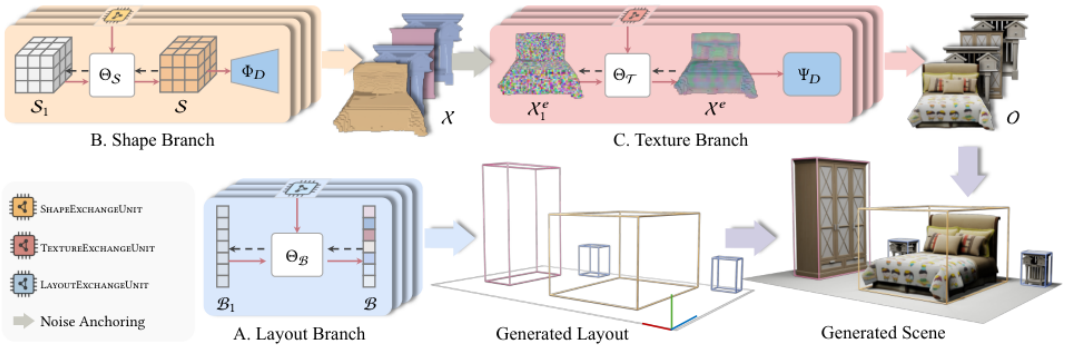

Figure 3: FlowScene Pipeline. Conditioned on $\mathcal{G}_M$ in Figure 2, FlowScene stacks three coordinated branches to generate a scene. *A. Layout Branch* generates the scene layout from noise $\mathcal{B}_1$, where the x-axis (red) points to the right and the z-axis (blue) points to the front. The LayoutExchangeUnit iteratively applies temporal layout constraints to the generation process by conditioning $\Theta_{\mathcal{B}}$. *B. Shape Branch* is parallel to the layout branch. It exchanges shape information among nodes according to $\mathcal{G}_M$ through the ShapeExchangeUnit. At inference, we sample noise $\mathcal{S}_1$, and generate latent shape codes $\mathcal{S}$ via $\Theta_{\mathcal{S}}$, and decode them into voxelized shapes $\mathcal{X}$ using Φ_D. *C. Texture Branch* is subsequent to the shape branch. It anchors Gaussian noise onto $\mathcal{X}$ to form $\mathcal{X}_1^e$, which is then denoised by $\Theta_{\mathcal{T}}$ into structured latents $\mathcal{X}^e$. Finally, Ψ_D decodes them into objects $\mathcal{O}$, which are scaled and placed into the generated layout $\mathcal{B}$ to complete the scene.

Figure 3: Illustrates the complete FlowScene pipeline with three coordinated branches (layout, shape, texture) and their respective exchange units that enable collaborative reasoning across the multimodal graph.

Limitations

- Current implementation focuses primarily on synthetic indoor environments and has not been evaluated on outdoor scenes
- Generation quality is intrinsically dependent on the precision of input multimodal graph construction
- Missing key spatial relationship edges in the graph may reduce layout constraints and result in object interpenetration
- Model performance may degrade when input images are captured from limited or biased viewpoints